Demand Planning UI: how it works

Oct 23, 2026 · @Tiyashi Chatterjee

Purpose

This tool predicts how many Eveready products will sell each week and each month, and checks the sales team's forecasts against budget, last year and stock. This page explains how each part works.

Data behind the screens

What we forecast: the units Eveready sells to its distributors each week (called primary sales), for each product, for the whole country and for each Kolkata depot.

The pipeline reads three files:

File	What it holds	Used by
`eveready_mock_data_v2.csv` (91 columns, 26,934 rows)	Weekly sales, sales-team forecasts at 3, 2 and 1 months ahead, budget, orders, stock, quick commerce, lost sales, transfers, MOQ columns	Every screen
`ext.csv` (external factors)	Weekly weather, power cuts, heatwave and cyclone flags, dengue, zinc price, rural income, LED price, expected values, festival labels	Forecast features, Weather screen, driver panels
`competitor.csv` (Nielsen-style)	Retail sales, distribution and promotion for 35 competitor brands, including 8 liquid vaporizer brands	Not used by any screen yet; reserved for new product forecasting

Coverage: 157 weeks (Oct 2023 to Sep 2026), 30 products, the whole country, and five Kolkata depots (Barasat, Barrackpore, Baruipur, Dankuni, Taratala). The depots are only about 7% of national sales, so depot screens describe Kolkata, not India.

Some columns are blank on purpose. For example, quick-commerce columns are filled only for the 16 products sold there, and four new products start selling later than the rest.

The screens

The tool has four screens: Forecast & accuracy, Forecast challenge, Quick commerce and Weather & shocks. The browser address holds the screen and tab (for example `#/forecast?view=monthly`), so refresh, back and shared links return to the same place.

Controls shared by every screen

Control	Options	Notes
Category, Product	All, or one category or product	Changing the category resets the product
Level	National, Kolkata depots	Quick commerce and Weather are national only
Measure	Units, Value (₹)	Hidden on the Weather screen
Light / dark	Toggle in the top bar	Charts re-read their colours on change
Scenario controls	Button on the Forecast screen	Described below

Forecast & accuracy

The weekly outlook is the main view. It shows every week up to March 2027: actuals up to the last week of data, then the model forecast only from 5 Oct 2026. Headline tiles sit above a chart with these lines: Actual, Forecast (ensemble), P50 (median), Sales-team forecast (3 months ahead), Last year, Budget, the Model M3, M2 and M1 runs, and Supply-constrained (it appears only when supply availability is set below 100%). Budget and the M3, M2 and M1 runs are hidden when the screen opens; click a legend entry to show them. A Table view lists the same numbers by week.

Clicking a week opens a panel that explains it: what each model says (seasonal naive, ETS and LightGBM, each as a bar showing how far it pulls the forecast from the starting point, times its weight in the blend), what was going on that week, and how the week compares with last month or the same month last year. When ETS is blank for a week, the panel lists only the models that are active.

The Monthly runs tab shows each target month as the M3, M2 and M1 runs made it, against actuals and the sales team, with a Target month selector. Under it, the accuracy card scores the runs against the industry reference and Eveready's own figure (see How accuracy is measured). The monthly tab does not apply scenario controls.

Scenario controls

A side panel with three levers and four presets. Nothing changes on the page until Apply is pressed (draft, then apply), and Reset returns to the base case.

Lever	Range	What it does
Overall demand	−20% to +20%	Moves every forecast week by the same percentage
Festive weeks	−30% to +30%	An extra change in weeks that fall in a festival or its sell-in period
Supply availability	80% to 100%	Share of demand the plants can ship; draws a separate dotted line and leaves demand unchanged

Presets: Base case, Weak festive season (−3% overall, −15% festive), Strong festive season (+2% overall, +15% festive) and Supply squeeze (88% availability). The levers scale the forecast weeks of the Forecast and P50 lines. The models are not re-run, so a scenario is an overlay and not a new forecast.

Forecast challenge

Every sales forecast for a chosen target month and lead (M3, M2 or M1) is checked against four references: budget, last year, the model, and current depot stock. A forecast is flagged when enough checks fail. The rules are adjustable: budget band (±20% to ±60%), last-year band (±25% to ±100%), model band (±25% to ±75%), how many of the three reference checks must fail (1 to 3), and maximum stock cover (6 to 16 weeks). The table can show Flagged forecasts only or All, and a note counts the flags raised by type.

Quick commerce & OTIF

National only. It compares what quick-commerce partners ordered against what they actually sold to shoppers (their sell-out), shows delivery performance (OTIF against fulfilment, which differ because OTIF scores whole order lines), and plans the next 13 weeks of shipments given the stock partners hold. Controls: a window of 13 weeks, 26 weeks or all, and the buffer stock a partner keeps (1, 2 or 3 weeks). A By product table closes the screen.

Weather & shocks

National only. For a chosen monsoon season (2024, 2025 or 2026), it shows demand against the forecast made just before the weather event, and actual rain against expected rain, as a Chart or a Table, so a shock and its recovery can be read week by week.

How the forecast is made

The forecast blends three models. Each one predicts every product's sales for each week, up to 26 weeks ahead, and the blend averages them using fixed weights. If one model has no value for a week, the others share its weight.

Model	Weight (country)	Weight (depots)	In plain words
Seasonal naive	20%	30%	Same week last year, adjusted for how sales are running now
Damped ETS	30%	0%	Smooths the sales history to find the level, trend and yearly pattern; needs two years of history, so it runs nationally only
LightGBM	50%	70%	A machine-learning model that learns from recent sales, last year, the calendar, price, weather and festivals

Festivals. Durga Puja and Diwali fall on different dates each year, so "the same week last year" often misses them. Around festivals, the models compare with last year's festival week instead, and LightGBM is told how many weeks away the nearest festival is.

Training. LightGBM learns from past sales. It predicts the change from a recent level, and odd single weeks are given less weight so they do not distort it.

M3, M2 and M1. At each month-end the models are retrained on everything known so far. The forecast made 3, 2 or 1 months before a month is that month's M3, M2 or M1 forecast.

P50. The median forecast: half of outcomes are expected above it and half below. It is set from how far off the model was in past tests.

Scenarios are applied afterwards, as a percentage overlay on the forecast weeks. They do not retrain anything.

How accuracy is measured and read

Accuracy is 1 minus (total forecast error divided by total actual sales), so 90% means the forecast was off by 10% of sales. The accuracy card adds all products together, then scores the M3, M2 and M1 forecasts over the ten completed months (Dec 2025 to Sep 2026). It also shows over-forecasts and under-forecasts separately, as the CFO asked, because they can cancel each other out inside a total.

Three reference points sit on the card: the industry reference (about 80% at M3 rising to 90% at M1, with M2 taken as the midpoint, 85%), Eveready's own figure quoted by the CFO (about 68% at M3), and the sales team's own forecast, scored the same way.

Weeks into months. A week can straddle two months, so each week is split by days. A week with 4 days in one month and 3 in the next counts 4/7 and 3/7.

How the data is rebuilt

The screens read ready-made data files. Scripts in the project's pipeline folder (`api/pipeline/`) produce them, run in this order:

1. `run_clean.py` for National and for Kolkata depot: backtests the ensemble from week 104 and makes the final forecast (about 3 minutes nationally, 2 for depots).
2. `run_band.py` for each level: measures backtest errors at 13, 19 and 26 weeks ahead, used for the P50 line and the range (about 2.5 and 1.8 minutes).
3. `build_vintages.py` for each level: retrains at each month-end and writes the M3, M2 and M1 runs (about 1.6 and 4 minutes). The two levels are then merged into `vintages.json` , and `add_hist.py` adds monthly actuals back to Nov 2023.
4. `build_weather.py` , `build_ui_data.py` , `build_challenge.py` , `build_qc.py` and `build_drivers.py` : the page data for each screen.
5. Copy `ui_data.json` , `vintages.json` , `challenge.json` , `weather.json` , `qc.json` and `drivers.json` into `api/data/` .